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Speech and Language Processing

Chapter 6:
Vector Semantics

Vector semantics is a way of representing the meaning of language as numerical vectors so that computers can mathematically compare relationships between meanings.

Let's define words by their usages

In particular, words are defined by their environments (the words around them)

Zellig Harris (1954): If A and B have almost identical environments we say that they are synonyms.

What does ong choi mean?

Suppose you see these sentences:

- Ongchoi is delicious **sautéed with garlic**.
- Ongchoi is superb **over rice**
- Ongchoi **leaves** with salty sauces

And you've also seen these:

- ...spinach **sautéed with garlic over rice**
- Chard stems and **leaves** are **delicious**
- Collard greens and other **salty** leafy greens

Conclusion:

- Ongchoi is a leafy green like spinach, chard, or collard greens

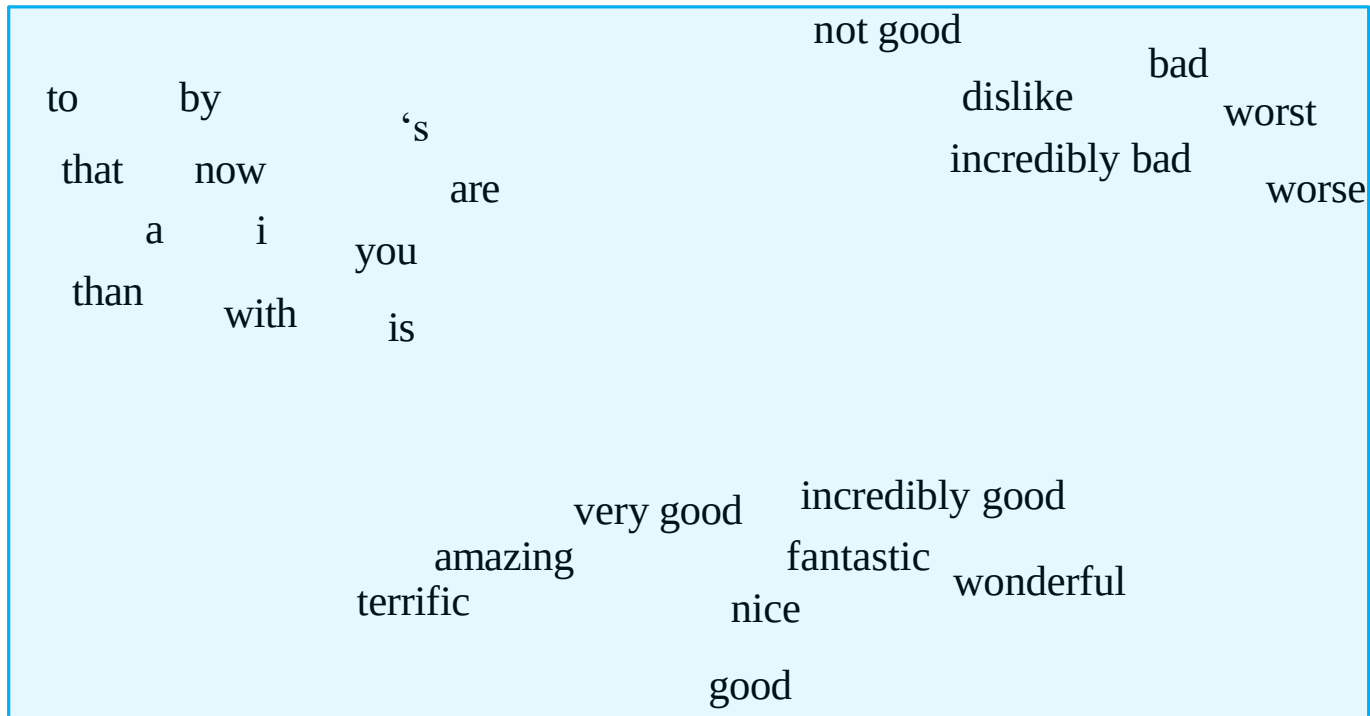
Ong choi: *Ipomoea aquatica* "Water Spinach"



Build a new model of meaning focusing on similarity

Each word = a vector

Similar words are "nearby in space"



Define a word as a vector

Called an "embedding" because it's embedded into a space

The standard way to represent meaning in NLP

Fine-grained model of meaning for similarity

- NLP tasks like sentiment analysis
 - With words, requires **same** word to be in training and test
 - With embeddings: ok if **similar** words occurred!!!
- Question answering, conversational agents, etc

2 kinds of embeddings

Tf-idf

- A common baseline model
- Sparse vectors
- Words are represented by a simple function of the counts of nearby words

Word2vec

- Dense vectors
- Representation is created by training a classifier to distinguish nearby and far-away words

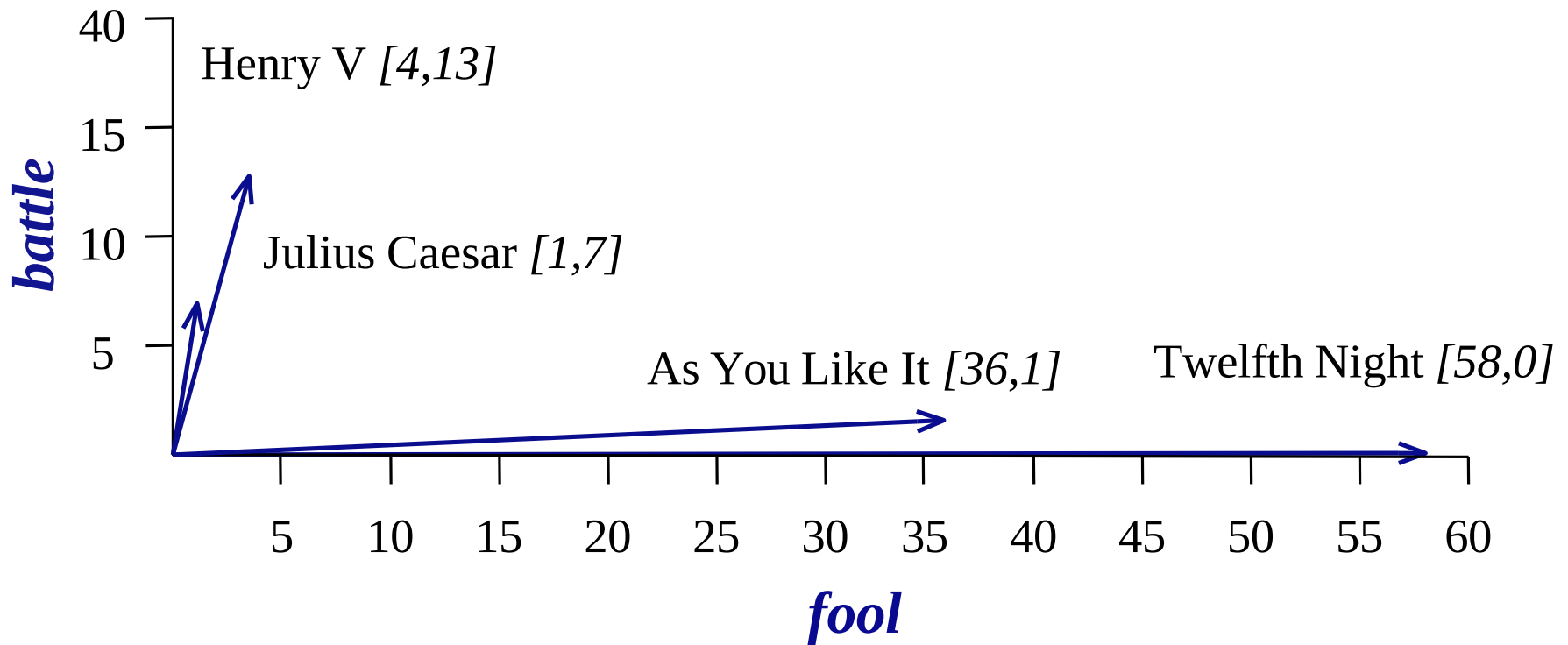
Review: words, vectors, and co-occurrence matrices

Term-document matrix

Each document is represented by a vector of words

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Visualizing document vectors



Vectors are the basis of information retrieval

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Vectors are similar for the two comedies
Different than the history

Comedies have more fools and wit and
fewer battles.

Words can be vectors too

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

battle is "the kind of word that occurs in Julius Caesar and Henry V"

fool is "the kind of word that occurs in comedies, especially Twelfth Night"

More common: word-word matrix (or "term-context matrix")

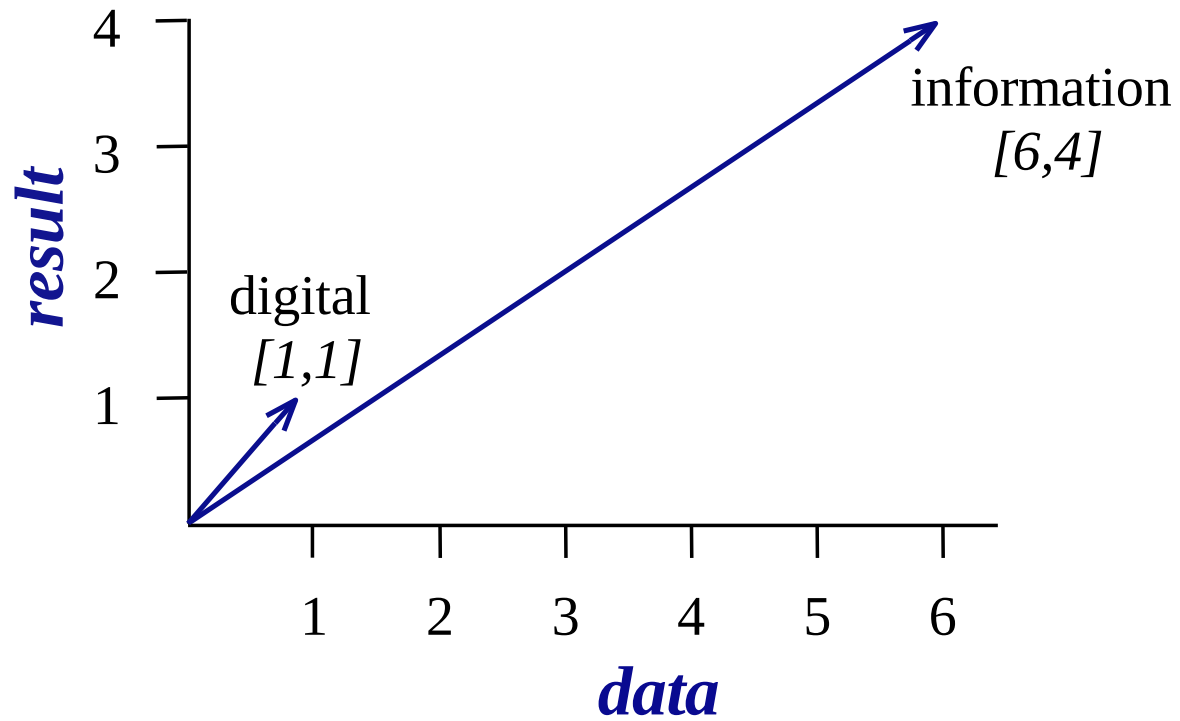
Two **words** are similar in meaning if their context vectors are similar

sugar, a sliced lemon, a tablespoonful of **apricot** jam, a pinch each of,
their enjoyment. Cautiously she sampled her first **pineapple** and another fruit whose taste she likened
well suited to programming on the digital **computer.** In finding the optimal R-stage policy from
for the purpose of gathering data and **information** necessary for the study authorized in the

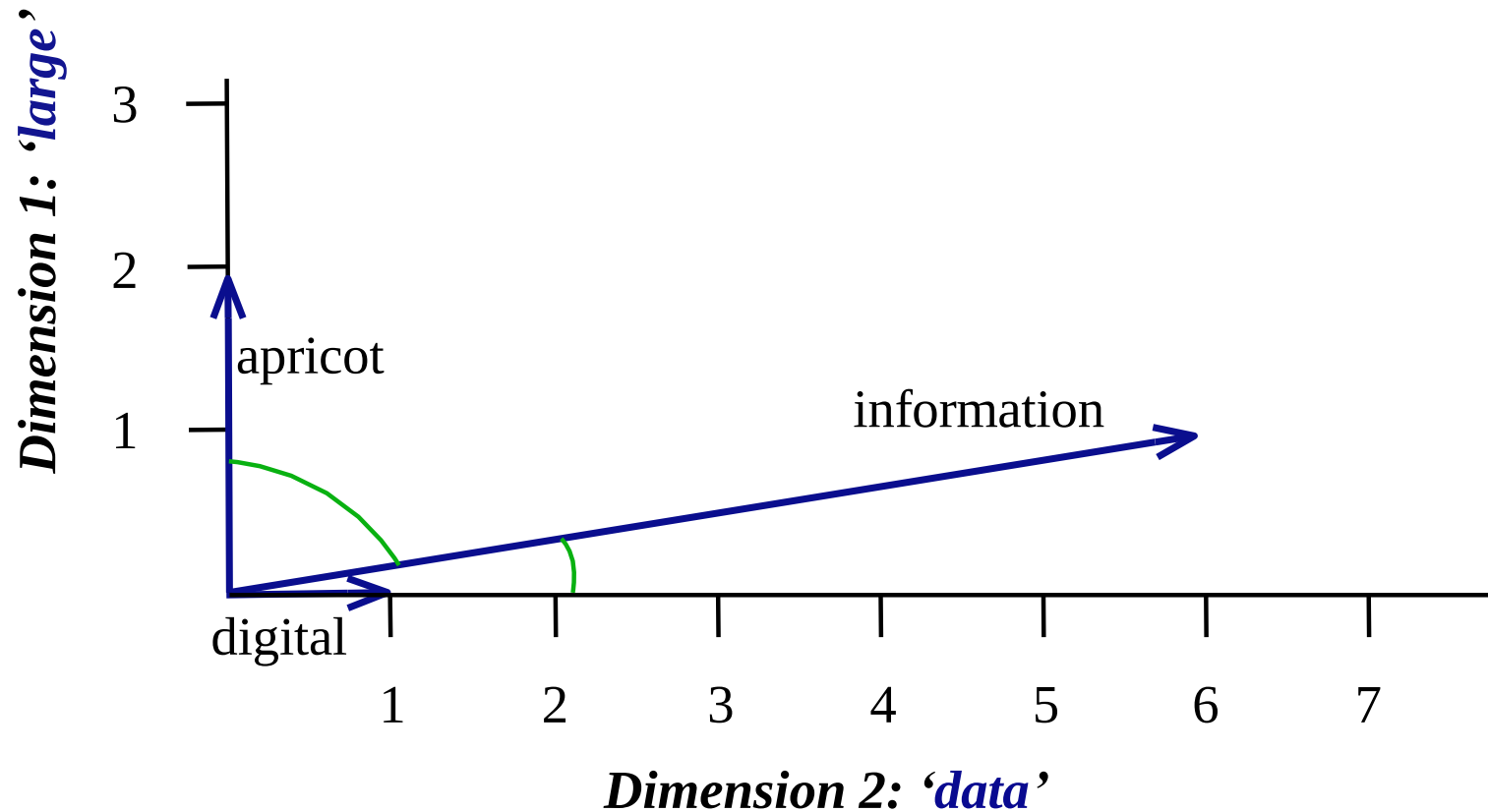
Context word

	aardvark	computer	data	pinch	result	sugar	...
apricot	0	0	0	1	0	1	
pineapple	0	0	0	1	0	1	
digital	0	2	1	0	1	0	
information	0	1	6	0	4	0	

Target word



Visualizing cosines (well, angles)



Cosine for computing similarity Sec. 6.3

$$\text{cosine}(\vec{v}, \vec{w}) = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}| |\vec{w}|} = \frac{\sum_{i=1}^N v_i w_i}{\sqrt{\sum_{i=1}^N v_i^2} \sqrt{\sum_{i=1}^N w_i^2}}$$

v_i is the count for word v in context i
 w_i is the count for word w in context i .

$$\begin{aligned} \vec{a} \cdot \vec{b} &= |\vec{a}| |\vec{b}| \cos \theta \\ \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} &= \cos \theta \end{aligned}$$

→ →

→ →

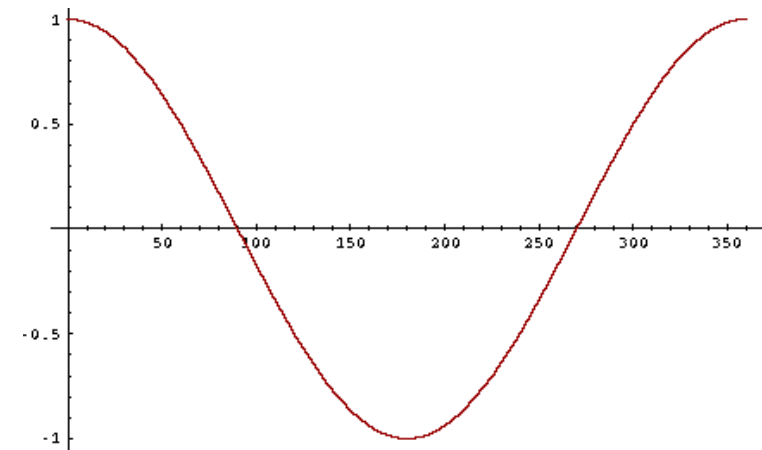
Cos(v, w) is the cosine similarity of v and w

Cosine as a similarity metric

-1: vectors point in opposite directions

+1: vectors point in same directions

0: vectors are orthogonal



Frequency is non-negative, so cosine range 0-1

$$\cos(\vec{v}, \vec{w}) = \frac{\vec{v} \bullet \vec{w}}{|\vec{v}| |\vec{w}|} = \frac{\vec{v}}{|\vec{v}|} \bullet \frac{\vec{w}}{|\vec{w}|} = \frac{\sum_{i=1}^N v_i w_i}{\sqrt{\sum_{i=1}^N v_i^2} \sqrt{\sum_{i=1}^N w_i^2}}$$

Which pair of words is more similar?

cosine(apricot, information) =

$$\frac{1+0+0}{\sqrt{1+0+0} \sqrt{1+36+1}} = \frac{1}{\sqrt{38}} = .16$$

cosine(digital, information) =

$$\frac{0+6+2}{\sqrt{0+1+4} \sqrt{1+36+1}} = \frac{8}{\sqrt{38} \sqrt{5}} = .58.$$

cosine(apricot, digital) =

$$\frac{0+0+0}{\sqrt{1+0+0} \sqrt{0+1+4}} = 0$$

	large	data	computer
apricot	1	0	0
digital	0	1	2
information	1	6	1

But raw frequency is a bad representation

Frequency is clearly useful; if *sugar* appears a lot near *apricot*, that's useful information.

But overly frequent words like *the*, *it*, or *they* are not very informative about the context

Need a function that resolves this frequency paradox!

tf-idf: combine two factors

how important a word is in a document relative to a collection of documents.

tf: term frequency. frequency count (usually log-transformed):

$$\text{tf}_{t,d} = \begin{cases} 1 + \log_{10} \text{count}(t, d) & \text{if } \text{count}(t, d) > 0 \\ 0 & \text{otherwise} \end{cases}$$

Idf: inverse document frequency: tf-

$$\text{idf}_i = \log \left(\frac{N}{\text{df}_i} \right)$$

Total # of docs in collection

Words like "the" or "good" have very low idf

of docs that have word i

tf-idf value for word t in document d:

$$w_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$

Word	df	idf
Romeo	1	1.57
salad	2	1.27
Falstaff	4	0.967
forest	12	0.489
battle	21	0.074
fool	36	0.012
good	37	0
sweet	37	0

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	0.074	0	0.22	0.28
good	0	0	0	0
fool	0.019	0.021	0.0036	0.0083
wit	0.049	0.044	0.018	0.022

A tf-idf weighted term-document matrix for four words in four Shakespeare plays

Summary for Part I

- Idea of Embeddings: Represent a word as a function of its distribution with other words
- Tf-idf
- Cosines